

Vishal Kumar Rai

Research Associate, Centre for Social and Economic Progress (CSEP)

Ph.D. (Thesis Submitted), Economics, Indian Statistical Institute – Delhi (ISI-D)

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EDUCATION

Ph.D. in Economics	2026
Indian Statistical Institute – Delhi	
Status: Thesis submitted	
M.A. in Economics	2019
Delhi School of Economics	
B.A. (Hons.) Business Economics	2015
University of Delhi	

PROFESSIONAL EXPERIENCE

Research Associate	Aug 2026 – Present
Centre for Social and Economic Progress (CSEP)	
Vertical: Growth, Finance, Health and Human Development	

DISSERTATION RESEARCH

Essays on Life-Cycle Decisions under Environmental and Epidemiological Shocks

Supervisor: Prof. Monisankar Bishnu

Chapter 1: Climate, Pollution and Endogenous Retirement

Examines how environmental risk shapes life-cycle consumption, health expenditure, and retirement decisions, showing that pollution alters the link between health investment and retirement timing.

Chapter 2: Endogenous Retirement in the Presence of a Pandemic

Studies life-cycle behavior under pandemic risk, showing that pandemic shocks generate heterogeneous health and labor-supply responses across age cohorts.

Chapter 3: Occupational Variation in Work Strain: A Life-Cycle Model of Retirement

Introduces occupational heterogeneity in work strain into a life-cycle framework, showing how working conditions interact with health expenditure and survival to shape retirement behavior.

RESEARCH PROJECTS

Health, Endogenous Retirement and Demand for Annuity

Department of Science and Technology (DST), Government of India

Principal Investigator: Prof. Monisankar Bishnu

Contributed to the development, calibration, and numerical implementation of a life-cycle model of health, ageing, retirement, and annuity demand. Constructed occupation-specific wage, mortality, survival, and frailty profiles using PLFS, LASI, SRS, CDC, and NOMS data; estimated model parameters, implemented numerical optimization in Julia, and conducted policy simulations under alternative pension and tax regimes.

WORKING PAPERS

A Matter of Timing: Health and Economic Responses to Pandemic Shocks over the Life Cycle

Under review at the *Journal of Health Economics*.

CONFERENCE PRESENTATIONS

Endogenous Retirement in the Presence of a Pandemic

Presented at the Macroeconomics Session, CDE–IEDS International Conference on *Economic Development, Demographic Transition and Environmental Sustainability*, Delhi School of Economics, New Delhi, June 2026.

COMPUTATIONAL TOOLS

SwitchingControl.jl

Registered Julia package for solving dynamic systems with endogenous regime switching. Built on SciMLBase and DifferentialEquations.jl using ContinuousCallback event detection to transition between distinct pre- and post-switch trajectory dynamics.

SKILLS

Programming: Julia (Scientific Computing, Numerical Optimization), Python

Methods: Structural Modeling, Calibration, Dynamic Programming, Numerical Methods

Data (India): LASI, SRS, PLFS

Data (US): MEPS, CPS, CDC Life Tables, NOMS

Tools: L^AT_EX, GitHub

RESEARCH INTERESTS

Labor Economics • Health Economics • Climate Economics • Structural Macroeconomics • Life-Cycle Modeling • Ageing and Retirement

TEACHING EXPERIENCE

Teaching Assistant, Indian Statistical Institute – Delhi

Prof. Chetan Ghate

Macroeconomics I and Global Macroeconomics for M.S. Quantitative Economics and Ph.D. students; conducted tutorials and assisted with assessment and course coordination.

REFERENCES

Monisankar Bishnu (Supervisor)

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